# CHEM 1301 Practice Exam 3-2

**Chapters 7, 8, and 9 — 35 multiple-choice questions**

**Instructions.** Choose the single best answer (a–e) for each question. You may use a periodic table. Solubility rules are **not** provided. Calculators are allowed. Answers are rounded to the precision shown. Time yourself: aim to finish in 75 minutes or less.

**Answer sheet.** Print the bubble answer sheet (`CHEM1301/exam-prep/answer-sheet/answer-sheet.pdf`, landscape, 100% scale) and fill in one circle per question, rows 1–35 (A–E match choices a–e). Leave the rest blank.

**1.** The reaction $HCl(aq) + KOH(aq) \rightarrow H_2O(l) + KCl(aq)$ occurs mainly because of which driving force?

a) Formation of water
b) Transfer of electrons
c) An increase in the number of ions in solution
d) Formation of a solid
e) Formation of a gas

**2.** On the basis of the general solubility rules, which compound is **insoluble** in water?

a) $K_2CO_3$
b) $BaSO_4$
c) $NH_4NO_3$
d) $CaCl_2$
e) $Na_3PO_4$

**3.** Which of the following compounds is expected to be **soluble** in water?

a) $PbSO_4$
b) $AgCl$
c) $Ca_3(PO_4)_2$
d) $Fe(OH)_3$
e) $K_2S$

**4.** Aqueous solutions of lead(II) nitrate, $Pb(NO_3)_2$, and sodium sulfate, $Na_2SO_4$, are mixed. What is the precipitate?

a) $PbSO_4$
b) $NaPb$
c) $NaNO_3$
d) $Pb(SO_4)_2$
e) No precipitate forms

**5.** Which is the correct **net ionic equation** for the reaction of aqueous barium chloride with aqueous potassium sulfate?

a) $K^+(aq) + Cl^-(aq) \rightarrow KCl(s)$
b) $Ba^{2+}(aq) + SO_4{}^{2-}(aq) \rightarrow BaSO_4(s)$
c) $Ba^{2+}(aq) + SO_4{}^{2-}(aq) \rightarrow BaSO_4(aq)$
d) $Ba^{2+}(aq) + 2Cl^-(aq) + 2K^+(aq) + SO_4{}^{2-}(aq) \rightarrow BaSO_4(s) + 2K^+(aq) + 2Cl^-(aq)$
e) $BaCl_2(aq) + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KCl(aq)$

**6.** When the equation below is balanced with the smallest whole-number coefficients, what is the **sum of all the coefficients**?

___ $H_2SO_4(aq)$ + ___ $Al(OH)_3(s) \rightarrow$ ___ $Al_2(SO_4)_3(aq)$ + ___ $H_2O(l)$

a) 4
b) 6
c) 8
d) 11
e) 12

**7.** In the reaction $K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow BaCrO_4(s) + 2KNO_3(aq)$, the spectator ions are:

a) $NO_3{}^-$ and $CrO_4{}^{2-}$
b) $Ba^{2+}$ and $CrO_4{}^{2-}$
c) There are no spectator ions
d) $K^+$ and $Ba^{2+}$
e) $K^+$ and $NO_3{}^-$

**8.** What are the products when aqueous nitric acid reacts with aqueous potassium hydroxide?

a) $KNO_2(aq)$ and $H_2O(l)$
b) $KH(aq)$ and $NO_3OH(aq)$
c) $K_2O(s)$ and $HNO_3(aq)$
d) $KNO_3(aq)$ and $H_2O(l)$
e) $KNO_3(s)$ and $H_2O(l)$

**9.** Which pair contains **only** the most common strong bases named in this course?

a) HCl and NaOH
b) $Mg(OH)_2$ and $Fe(OH)_3$
c) KOH and $HNO_3$
d) $Ca(OH)_2$ and $Al(OH)_3$
e) NaOH and KOH

**10.** An aqueous solution of barium nitrate, $Ba(NO_3)_2(aq)$, contains:

a) $Ba^{2+}$, $N^{3-}$, and $O^{2-}$ ions
b) separated $Ba^{2+}$ and $NO_3{}^-$ ions, two $NO_3{}^-$ for every $Ba^{2+}$
c) separated $Ba^{2+}$ and $NO_3{}^-$ ions, one $NO_3{}^-$ for every $Ba^{2+}$
d) neutral Ba atoms and $NO_3$ molecules
e) intact $Ba(NO_3)_2$ units

**11.** In the reaction $4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$, how many electrons does **each oxygen atom** gain?

a) 6
b) 4
c) 3
d) 2
e) 1

**12.** The reaction $2HCl(aq) + Na_2CO_3(aq) \rightarrow CO_2(g) + H_2O(l) + 2NaCl(aq)$ is best classified as:

a) gas evolution and acid–base
b) precipitation only
c) combustion
d) decomposition
e) oxidation–reduction and single-replacement

**13.** When heated, potassium chlorate reacts: $2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$. This is best classified as a:

a) synthesis reaction
b) acid–base reaction
c) double-displacement reaction
d) decomposition reaction
e) combustion reaction

**14.** The average mass of a single carbon atom is:

a) 12.01 amu
b) 1 amu
c) $6.022 \times 10^{23}$ g
d) $1.66 \times 10^{-24}$ amu
e) 12.01 g

**15.** Which 10.0 g sample contains the **greatest number of atoms**?

a) gold (196.97 g/mol)
b) boron (10.81 g/mol)
c) aluminum (26.98 g/mol)
d) iron (55.85 g/mol)
e) xenon (131.3 g/mol)

**16.** How many moles of iron are in 25.0 g of Fe?

a) 0.448 mol
b) 2.23 mol
c) 4.48 mol
d) 1396 mol
e) $2.70 \times 10^{23}$ mol

**17.** How many atoms are in 2.75 mol of calcium?

a) $4.57 \times 10^{-24}$ atoms
b) $4.13 \times 10^{22}$ atoms
c) $1.66 \times 10^{23}$ atoms
d) $1.66 \times 10^{24}$ atoms
e) $6.64 \times 10^{25}$ atoms

**18.** What is the mass of $5.00 \times 10^{21}$ atoms of copper?

a) $1.31 \times 10^{-4}$ g
b) $8.30 \times 10^{-3}$ g
c) 0.528 g
d) $7.65 \times 10^{3}$ g
e) $3.18 \times 10^{23}$ g

**19.** What is the molar mass of calcium phosphate, $Ca_3(PO_4)_2$?

a) 135.1 g/mol
b) 215.2 g/mol
c) 230.0 g/mol
d) 246.2 g/mol
e) 310.2 g/mol

**20.** What is the mass of 0.350 mol of sulfuric acid, $H_2SO_4$?

a) $3.57 \times 10^{-3}$ g
b) 11.2 g
c) 28.7 g
d) 34.3 g
e) 280 g

**21.** How many moles of nitrogen dioxide, $NO_2$, are in 125 mg of the gas?

a) $2.72 \times 10^{-3}$ mol
b) $4.17 \times 10^{-3}$ mol
c) 2.72 mol
d) 5.75 mol
e) $1.64 \times 10^{21}$ mol

**22.** What is the mass percent of nitrogen in ammonium sulfate, $(NH_4)_2SO_4$?

a) 10.60%
b) 13.33%
c) 21.20%
d) 24.27%
e) 48.43%

**23.** Which pair of compounds has the **same empirical formula**?

a) CO and $CO_2$
b) $C_2H_4$ and $C_2H_6$
c) $C_2H_2$ and $C_6H_6$
d) $H_2O$ and $H_2O_2$
e) NO and $NO_2$

**24.** An iron oxide is 69.94% Fe and 30.06% O by mass. What is its empirical formula?

a) $FeO_2$
b) $Fe_3O_4$
c) FeO
d) $Fe_3O_2$
e) $Fe_2O_3$

**25.** A compound has the empirical formula $NO_2$ and a molar mass of 92.02 g/mol. What is its molecular formula?

a) $NO_4$
b) $NO_2$
c) $N_4O_8$
d) $N_2O_4$
e) $N_3O_6$

**26.** In **combustion analysis**, a carbon-containing sample is:

a) burned in excess oxygen so it fully converts to products, giving the percent C (and other elements) and so the empirical formula
b) heated with an electric current until it decomposes into its elements
c) weighed to determine its molecular formula without any other data
d) burned in a limited amount of oxygen to measure its molar mass directly
e) dissolved in water to see whether it conducts electricity

**27.** Why must a chemical equation be balanced before it is used in a stoichiometry calculation?

a) Balancing is needed only so the molar masses can be calculated
b) The mole ratios used in the calculation come from the coefficients of the balanced equation
c) Balancing converts the coefficients from grams into moles
d) Balancing makes the total mass of the reactants equal the mass of each product
e) An unbalanced equation has no limiting reactant

**28.** For $2Al_2O_3 \rightarrow 4Al + 3O_2$, how many molecules of $O_2$ are formed from 6 formula units of $Al_2O_3$?

a) 3
b) 4
c) 9
d) 12
e) 18

**29.** Ethane burns according to $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. How many moles of $CO_2$ form when 0.750 mol of $O_2$ reacts completely?

a) 0.214 mol
b) 0.429 mol
c) 0.750 mol
d) 1.31 mol
e) 1.50 mol

**30.** According to the course, one reason the actual yield of a reaction is usually **less** than the theoretical yield is:

a) mass is not conserved in most chemical reactions
b) the excess reactant also turns into product
c) side reactions consume some of the reactants or products
d) the limiting reactant is never used up
e) the theoretical yield is calculated from the excess reactant

**31.** For $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$, what mass of $K_2SO_4$ is produced from 25.0 g of KOH with excess $H_2SO_4$?

a) 0.223 g
b) 21.9 g
c) 25.0 g
d) 38.8 g
e) 77.6 g

**32.** For $3Mg + N_2 \rightarrow Mg_3N_2$, 0.600 mol of Mg reacts with 0.250 mol of $N_2$. What mass of $Mg_3N_2$ can form?

a) 0.200 g
b) 20.2 g
c) 21.6 g
d) 25.2 g
e) 60.6 g

**33.** For $Pb(NO_3)_2 + 2NaCl \rightarrow PbCl_2 + 2NaNO_3$, what is the theoretical yield of $PbCl_2$ when 10.0 g of NaCl reacts with excess $Pb(NO_3)_2$?

a) 0.0856 g
b) 10.0 g
c) 23.8 g
d) 28.3 g
e) 47.6 g

**34.** For $N_2 + 3H_2 \rightarrow 2NH_3$, 14.0 g of $N_2$ reacts with 6.00 g of $H_2$, and 12.5 g of $NH_3$ is collected. What is the percent yield?

a) 37.0%
b) 62.5%
c) 73.4%
d) 89.3%
e) 136%

**35.** A student reports a percent yield of 112%. Which statement is correct?

a) The theoretical yield must have been calculated from the limiting reactant
b) The excess reactant was converted into additional product
c) Yields above 100% are normal whenever the reaction is fast
d) Side reactions produced extra product beyond the theoretical yield
e) Something is wrong, such as a wet or impure product or a calculation error, because the theoretical yield is the maximum possible